



```
[7]: import pandas as pd
data = {
    'Name' : ['Alice', 'Bob', 'Charlie', 'David'],
    'Age' : [34, 68, 75, 42],
    'Salary' : [50000, 35000, 90000, 25000]
}
df = pd.DataFrame(data)
print("Dataframe from scratch")
print(df)
df['Bonus']=df['Salary']*0.10
print("\nDataframe after adding 'Bonus' column")
print(df)
```

Dataframe from scratch

	Name	Age	Salary
0	Alice	34	50000
1	Bob	68	35000
2	Charlie	75	90000
3	David	42	25000

DataFrame after adding 'Bonus' column

	Name	Age	Salary	Bonus
0	Alice	34	50000	5000.0
1	Bob	68	35000	3500.0
2	Charlie	75	90000	9000.0
3	David	42	25000	2500.0

```
•[12]: df = pd.read_csv('experience.csv')
print("First 6 rows of the DataFrame")
print(df.head(6))
print("\nColumn names and data types: ")
df.info()
```

First 6 rows of the DataFrame

YearsExperience

0	1.1
1	1.3
2	1.5
3	2.0
4	2.2
5	2.9

Column names and data types:

<class 'pandas.DataFrame'>

RangeIndex: 9 entries, 0 to 8

Data columns (total 1 columns):

#	Column	Non-Null Count	Dtype
0	YearsExperience	9 non-null	float64

dtypes: float64(1)

memory usage: 204.0 bytes

```
[22]: import pandas as pd
df = pd.read_csv("values.csv")
print(df)
print("\nMissing values in DataFrame: ")
print(df)
print(df.isnull().sum())
df.Q1=df['Q1'].fillna(df['Q1'].mean())
print(df.Q1)
```

	Year	McDonalds_Revenue_\$Billion	Growth_rate_percent	Q1	Q2	Q3	Q4
0	1999	13.3	NaN	NaN	NaN	NaN	NaN
1	2000	14.2	7.0	NaN	NaN	NaN	NaN
2	2001	14.9	4.0	NaN	NaN	NaN	NaN
3	2002	15.4	4.0	NaN	NaN	NaN	3.0
4	2003	17.1	11.0	3.8	4.3	4.5	4.6
5	2004	18.6	8.0	4.4	4.7	4.9	4.5
6	2005	19.1	3.0	4.8	5.1	5.3	3.9
7	2006	20.9	9.0	4.9	5.4	5.5	5.1
8	2007	22.8	9.0	5.3	5.8	5.9	5.8

Missing values in DataFrame:

	Year	McDonalds_Revenue_\$Billion	Growth_rate_percent	Q1	Q2	Q3	Q4
0	1999	13.3	NaN	NaN	NaN	NaN	NaN
1	2000	14.2	7.0	NaN	NaN	NaN	NaN
2	2001	14.9	4.0	NaN	NaN	NaN	NaN
3	2002	15.4	4.0	NaN	NaN	NaN	3.0
4	2003	17.1	11.0	3.8	4.3	4.5	4.6
5	2004	18.6	8.0	4.4	4.7	4.9	4.5
6	2005	19.1	3.0	4.8	5.1	5.3	3.9
7	2006	20.9	9.0	4.9	5.4	5.5	5.1
8	2007	22.8	9.0	5.3	5.8	5.9	5.8
9	2008	23.5	3.0	5.6	6.1	6.3	5.6
10	2009	22.7	-3.0	5.1	5.6	6.0	6.0
11	2010	24.1	6.0	5.6	5.9	6.3	6.2
12	2011	27.0	12.0	6.1	6.9	7.2	6.8
13	2012	27.6	2.0	6.5	6.9	7.2	7.0
14	2013	28.1	2.0	6.6	7.1	7.3	7.1
15	2014	27.4	-2.0	6.7	7.2	7.0	6.6
16	2015	25.4	-7.0	6.0	6.5	6.6	6.3
17	2016	24.6	-3.0	5.9	6.3	6.4	6.0
18	2017	22.8	-7.0	5.7	6.0	5.8	5.3
19	2018	21.3	-7.0	5.1	5.4	5.4	5.4
20	2019	21.4	1.0	5.0	5.3	5.6	5.4
21	2020	19.2	-10.0	4.7	3.8	5.4	5.3
22	2021	23.2	21.0	5.1	5.9	6.2	6.0
23	2022	23.2	0.0	5.7	5.7	5.9	5.9

Year 0

McDonalds_Revenue_\$Billion 0

Growth_rate_percent 1

Q1 4

Q2 4

Q3 4

Q4 3

dtype: int64

0 5.43

1 5.43

2 5.43

```
0    5.43
1    5.43
2    5.43
3    5.43
4    3.80
5    4.40
6    4.80
7    4.90
8    5.30
9    5.60
10   5.10
11   5.60
12   6.10
13   6.50
14   6.60
15   6.70
16   6.00
17   5.90
18   5.70
19   5.10
20   5.00
21   4.70
22   5.10
23   5.70
```

```
Name: Q1, dtype: float64
```